

ITC C103 FUNDAMENTALS OF IT WW2 - M2U3 Midterms Assignment 5

BSIT - 102I

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Answers:

1. Before writing a program, developers often create a visual diagram to map out the sequence of events and decisions. What is this type of diagram called, and what is its purpose?

Answer: A flowchart is a visual diagram used to show the sequence of events, steps, and decisions in a program, helping developers plan and understand the program's logic.

2. Imagine you are building a website. When would you choose to use a compiled language for a complex backend system versus a scripting language for a simple web page, and how do their execution methods differ?

Answer: A compiled language is suitable for a complex backend system because it is compiled into machine code before execution, while a scripting language is suitable for simple web pages because its code is generally interpreted or executed at runtime.

3. In a program that manages user accounts, you need to store information like a user's name and email address. What are the two fundamental programming concepts used to store data, and what is the difference between them?

Answer: The two fundamental concepts are variables and constants. variables store data that can change during program execution, while constants store data that remains unchanged.

4. A popular application needs to greet users differently based on the time of day (morning, afternoon, or evening). What type of programming structure would you use to check the time and display the correct message?

Answer: A conditional structure, such as if, else if, and else, would be used to check the time and display the appropriate morning, afternoon, or evening greeting.

5. You need to process a list of 100 customer records to calculate their total order value. What programming concept would you use to automatically repeat the calculation for each record without writing the code 100 times?

Answer: You'd use a loop (example, a for loop) to run through the list of 100 records and apply the same calculation to each one, instead of writing the same code 100 times.

6. A team of developers is creating a large application. What is the difference between a procedure and a function, and in what real-life scenarios would a team use one over the other to write more organized and reusable code?

Answer: A procedure performs an action/task but doesn't return a value (example, printing a report).

A function performs a task and returns a value that can be used elsewhere (example, calculating a total).

Real-life use: Use a function when you need the result for further use (like calculating tax to add to a total). Use a procedure when you just need something done (like sending an email or displaying a message). Teams use both to keep code modular, reusable, and easier to maintain/debug.

7. Think about a real-world object like a car. What are three key components (properties, methods, and a class) that would define a "Car" object in a program?

Answer: Properties (attributes) - color, brand, speed, fuel level

Methods - (behaviors): accelerate, brake, honk

Class - the blueprint/template (Car) that defines what properties and methods every car object will have; each actual car (example, "myHonda") is an instance of that class.

8. You are an IT administrator for an organization using Windows 11. What are two common scripting tools you could use to automate tasks like managing user accounts or installing software on multiple computers?

Answer: PowerShell and Batch scripts (.bat), both let admins automate tasks like creating user accounts or installing/updating software across multiple machines. (PowerShell is more powerful and widely used for modern automation.)

9. Your company wants to launch a new productivity tool. Explain the main advantages and disadvantages of deploying it as a web application versus a locally installed application.

Answer: Web application is accessible from any device/browser, easier to update centrally, no installation needed. But it requires internet connection, potentially slower performance, dependent on browser compatibility. Locally installed application works offline, often faster/more responsive, can access system resources directly. But it must be installed/updated on each device, uses local storage/resources, harder to maintain across many machines